

Game Design Document (GDD)

Game Title

Final Bell

Version 1.0

1. Game Concept & Overview

Final Bell is a single-level survival horror game set inside a college dormitory during a zombie outbreak. The player takes the role of a student who must survive, explore the building, gather essential items, fight zombies and escape the dormitory before it is overrun.

The game focuses on atmospheric tension, resource management, and close-quarters combat, blending exploration with survival mechanics in a dark, horror-themed environment.

Genre

Survival/ Horror

2. Target Audience

- Age Group: **16+**
- Players who enjoy:
 - Horror and survival games
 - Exploration-based gameplay
 - Zombie-themed action
- Casual, core horror fans

Platform

PC (Windows)

3. Storyline / Theme

After the campus closes for the night and the final bell echoes through the halls, a deadly infection begins to spread. The dormitory becomes a trap filled with zombies. The player wakes up alone on the second floor of the dorm building and must:

- Search rooms for supplies
- Defend themselves using available weapons
- Find the key to unlock the main exit
- Escape the building to survive

The theme focuses on isolation, fear, and urgency with limited lighting and unpredictable enemy encounters.

4. Core Gameplay Mechanics

4.1 Player Movement

- Move: **W, A, S, D**
- Jump: **Space**
- Look: Mouse

4.2 Combat System

- Player can fight zombies using:
 - Bare hands
 - Bat
 - Axe
 - Knife
- Damage varies based on equipped weapon
- Zombies have health and can be killed

4.3 Health System

- Player health range: **0 – 100**
- Health decreases when attacked by zombies
- Health can be restored using:
 - Food (+10)
 - Energy drink (+15)
 - Injection (+40)
- When health reaches 0 -> **Game Over**

4.4 Inventory System

- Pickup items using Q
- Inventory stores:
 - Weapon
 - Consumables
 - Key
 - Torch
- Switch items using Tab key
- Player can switch to **Unarmed** state

4.5 Interaction System

- **E key:**
 - Open doors
 - Break glass case
 - Use key on main door

- Context – sensitive interactions

4.6 Torch / Flashlight System

- Toggle torch ON / OFF using F key
- Torch provides limited visibility in dark environments

5. Game Rules, Objectives & Progression

Objectives

- Explore the dormitory floors
- Collect weapons and health items
- Survive zombie encounters
- Locate the main exit door key
- Open the main exit door using the key
- Escape the building -> **Game Win**

Failure Condition

- Player health reaches 0 -> **Game Over Screen**

Progression

- Player starts in a safe room (2nd floor)
- Threat increases as player explores deeper areas
- Final challenge occurs near the main exit

6. Game World & Level Design

Main Menu Level

- UI only: Start / How to play / Credit / Quit

Level 1 – Dorm Building

- Floor 1
 - Main exit door
 - Cafeteria
 - Security room (Key location)
- Floor 2
 - Player room
 - Hallway rooms
 - Wide open area
 - Initial exploration

- Floor 3
 - Similar layout to 2nd floor for exploration

Navigation

- **NavMesh** enabled across floors and stairs

Level Design Principles

- Linear progression with optional exploration
- Strategic lighting to guide player
- Narrow hallways to enhance horror

7. Assets & Character

Environment Assets

- Dormitory rooms
- Hallways
- Staircases
- Cafeteria
- Doors and furniture

Character Assets

- Player character (student)
- Zombies
- Weapon models (bat, axe, knife)
- Props (food, medicine, key, torch)

Asset Sources

- Unreal Engine Starter Content
- Free asset platforms (Fab, Mixamo, Sketchfab)
- Custom Blueprint setups

8. UI / UX Design

User Interface Elements

- Health bar (progress bar)
- Main Menu:
 - Start Game
 - How to Play
 - Credit
 - Quit
- Pause Menu:
 - Resume
 - Restart
 - Quit
- Game Over Screen
 - Restart
 - Main Menu
- Game Win Screen
 - Restart
 - Main Menu

UX Design Considerations

- Clear feedback through messages and sounds
- Accessible controls and instructions

Audio Design

- Ambient loop (low volume) for constant tension
- Zombie sounds
- Player sounds

Credits / Attribution Plan

A “Credit” panel in the main menu lists:

- AI background: “Generated using ChatGPT (OpenAI)”

- Model sources: Mixamo, Sketchfab, Free3D
- Sound sources: Pixabay / Freesound

9. Lighting & Atmosphere

- Lighting system: Lumen
- Most lights set to Movable
- Low-intensity lighting with high contrast
- Flickering lights in key areas
- Post-process volume used for exposure control
- Dark horror atmosphere without sacrificing visibility

10. Technical Implementation (BluePrint)

Core Blueprints

- BP_ThirdPersonCharacter
- BP_Zombie
- BP_ZombieSpawner
- BP_ZombieAIController
- BP_FlickeringLight
- BP_ItemPickup
- BP_Door
- GM_MainMenu

Systems Implemented

- Health systems with UI binding
- Inventory and item switching
- Weapon equip system using sockets
- Zombie AI movement (AI Move TO + NavMesh)
- Door logic with key validation
- Game state management (Start, Pause, Game Over, Game Win)

Bug Fixing & Optimization

- Prevented multiple widget spawning
- Fixed input mode conflicts (UI / Game)
- Corrected collision and NavMesh issues
- Cleaned Blueprint execution flow
- Used variables and functions to reduce repetition

11. Future Improvements

- Advanced zombie AI with varied behaviors
- More levels and expanded building areas
- Improved sound design
- Save / load system for player progress
- Enhanced inventory UI